

torch.linalg.vander#

<https://pytorch.org/docs/stable/generated/torch.linalg.vander.html>

Description:

Generates a Vandermonde matrix. Returns the Vandermonde matrix V for $N > 1$. If $N = \text{None}$, then $N = x.\text{size}(-1)$ so that the output is a square matrix. Supports inputs of float, double, cfloat, cdouble, and integral dtypes. Also supports batches of vectors, and if x is a batch of vectors then the output has the same batch dimensions. Differences with `numpy.vander`:

Function Signature

`torch.linalg.vander(x, N=None) → Tensor#`

Parameters

Keyword Arguments

Parameters

Keyword

N (int, optional) – Number of columns in the output. Default: x.size(-1)

Code Examples

```
>>> x = torch.tensor([1, 2, 3, 5])
>>> linalg.vander(x)
tensor([[ 1,  1,  1,  1],
        [ 1,  2,  4,  8],
        [ 1,  3,  9, 27],
        [ 1,  5, 25, 125]])
>>> linalg.vander(x, N=3)
tensor([[ 1,  1,  1],
        [ 1,  2,  4],
        [ 1,  3,  9],
        [ 1,  5, 25]])
```

Detailed Documentation

Documentation

Generates a Vandermonde matrix.

Returns the Vandermonde matrix VV

for $N > 1$. If $N = \text{None}$, then $N = x.size(-1)$ so that the output is a square matrix.

Supports inputs of float, double, cfloat, cdouble, and integral dtypes. Also supports batches of vectors, and if x is a batch of vectors then the output has the same batch dimensions.

Differences with `numpy.vander`:

Unlike `numpy.vander`, this function returns the powers of x in ascending order. To get them in the reverse order call `linalg.vander(x, N).flip(-1)`.

x (Tensor) – tensor of shape $(*, n)$ where $*$ is zero or more batch dimensions consisting of vectors.

N (int, optional) – Number of columns in the output. Default: $x.size(-1)$